

R Assignment : II

Advanced Data Visualization and Statistical Analysis

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This assignment involves the advanced data visualization techniques, non-parametric tests, and linear regression diagnostics using R language. Below are the detailed, analysis and interpretations for each part of the code with their plotted results and analysis:

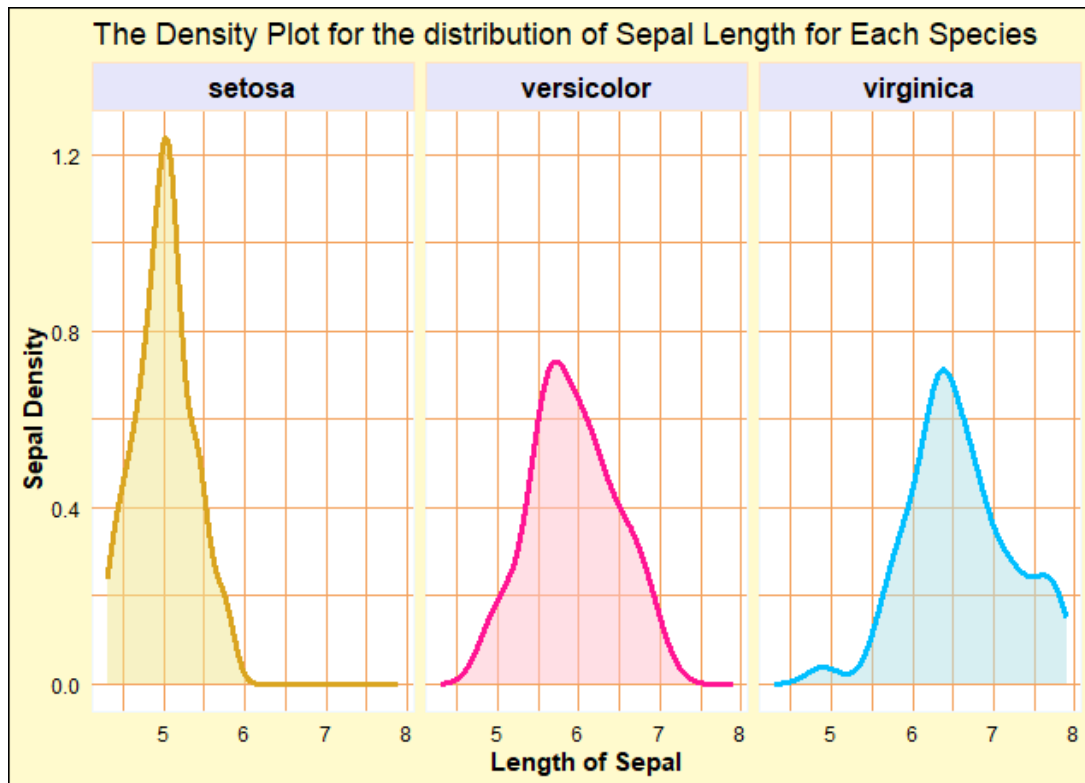
Part 1: Advanced Data Visualization

1. Density Plot with Facets

Objective: The main objective of part 1 to visualize the distribution of Sepal.Length for each species in the iris dataset.

Analysis of Results and Interpretation:

- **Setosa:** The density plot shows a very sharp peak, which indicates that the values of Sepal.Length are clustered in a narrow range. This reflects low variability in Sepal.Length among Setosa flowers, meaning they are relatively uniform in this characteristic.
- **Versicolor:** The graph has a wider peak compared to Setosa, indicating greater variability in Sepal.Length. The spread indicates that Sepal.Length values vary more for Versicolor species compared to Setosa.
- **Virginica:** The density curve is the widest and flattest of the three species. This means that Sepal.Length has the highest variability for Virginica, with values spread over a larger range.



2. Combined Scatter and Smooth Line Plot

Objective: To examine the relationship between hp (Horsepower) and mpg (Miles Per Gallon) in the mtcars dataset.

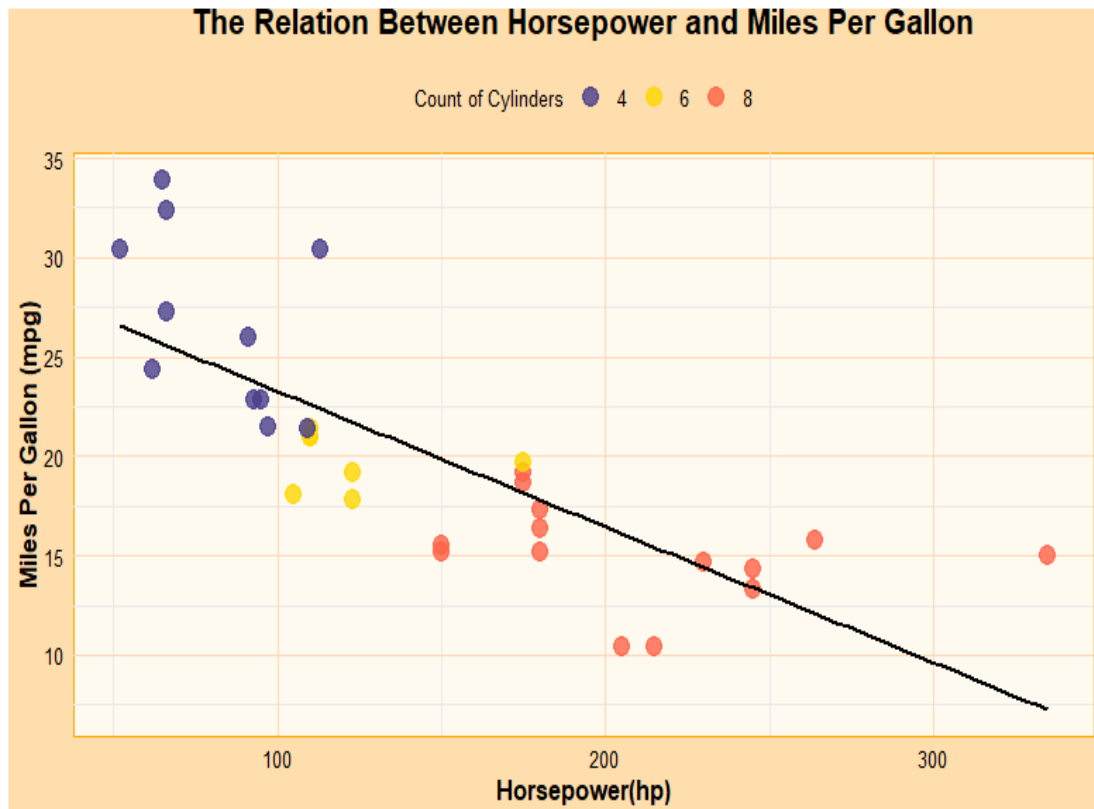
Analysis of Results and Interpretation:

- **Horsepower vs. Fuel Efficiency:** The scatterplot and smooth trend line suggest that higher horsepower cars (hp) are generally associated with lower fuel efficiency (mpg). This represents an inverse relationship between engine power and fuel consumption.

Effect of Number of Cylinders

- **8-cylinder vehicles:** The majority of the cars cluster at lower values of mpg, meaning that they consume more fuel.
- **4-cylinder vehicles:** The cars group at higher values of mpg, indicating a better fuel efficiency.

- **6-cylinder** vehicles are in the middle and intermediate efficiencies.
- **Conclusion:** Horsepower is a key predictor of mpg, and cylinder count helps explain further variation in mpg, offering insights into consumption patterns between different classes of vehicles.



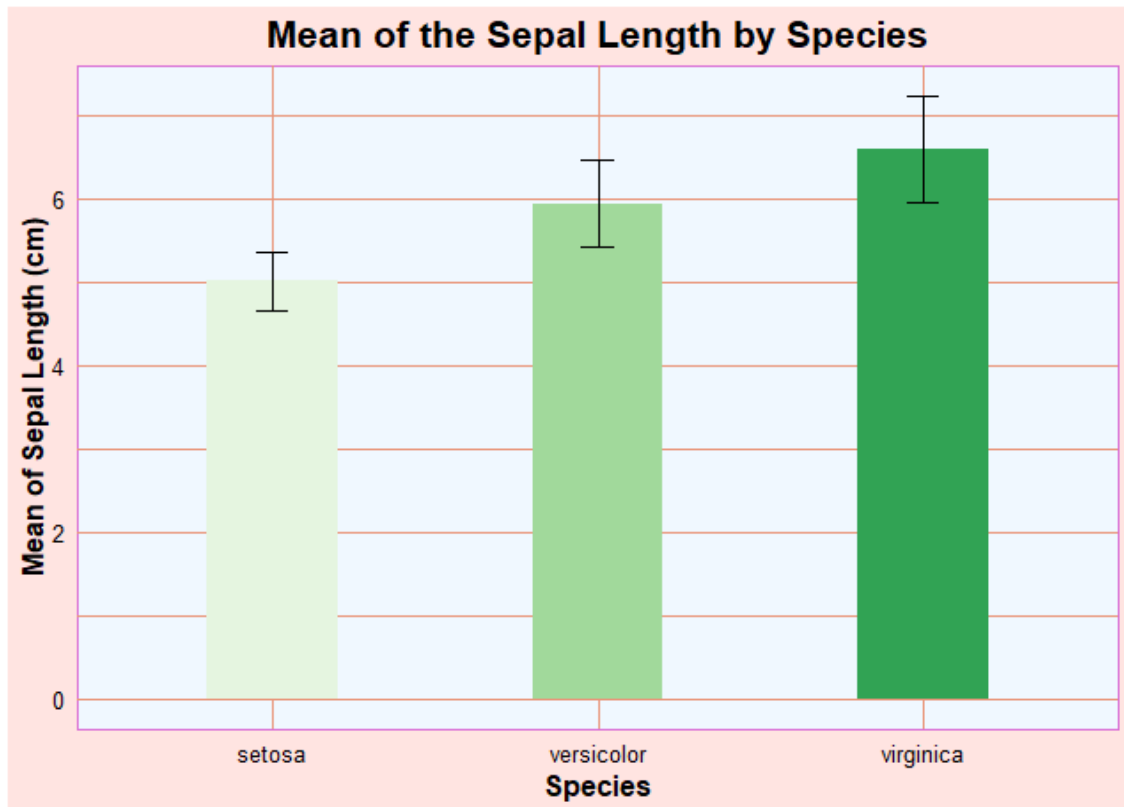
3. Bar Plot with Summary Statistics

Objective: To visualize the mean and standard deviation of Sepal.Length for each species in the iris dataset.

Analysis of Results and Interpretation:

- **Setosa:** Sepal.Length has the lowest mean and smallest standard deviation, meaning that this species is very consistent in the length of its sepals.
- **Versicolor:** The mean Sepal.Length is greater than Setosa, and the SD is average, indicating a better spread of sizes in this species.

- **Virginica:** This is the one with the highest mean Sepal.Length and the largest standard deviation of all, so the most variation in sepal length.
- **Conclusion:** The bar plot effectively expresses the difference in average lengths of sepals and variance in a species, thus giving an easy comparative understanding of them.



Part 2: Non-Parametric Tests

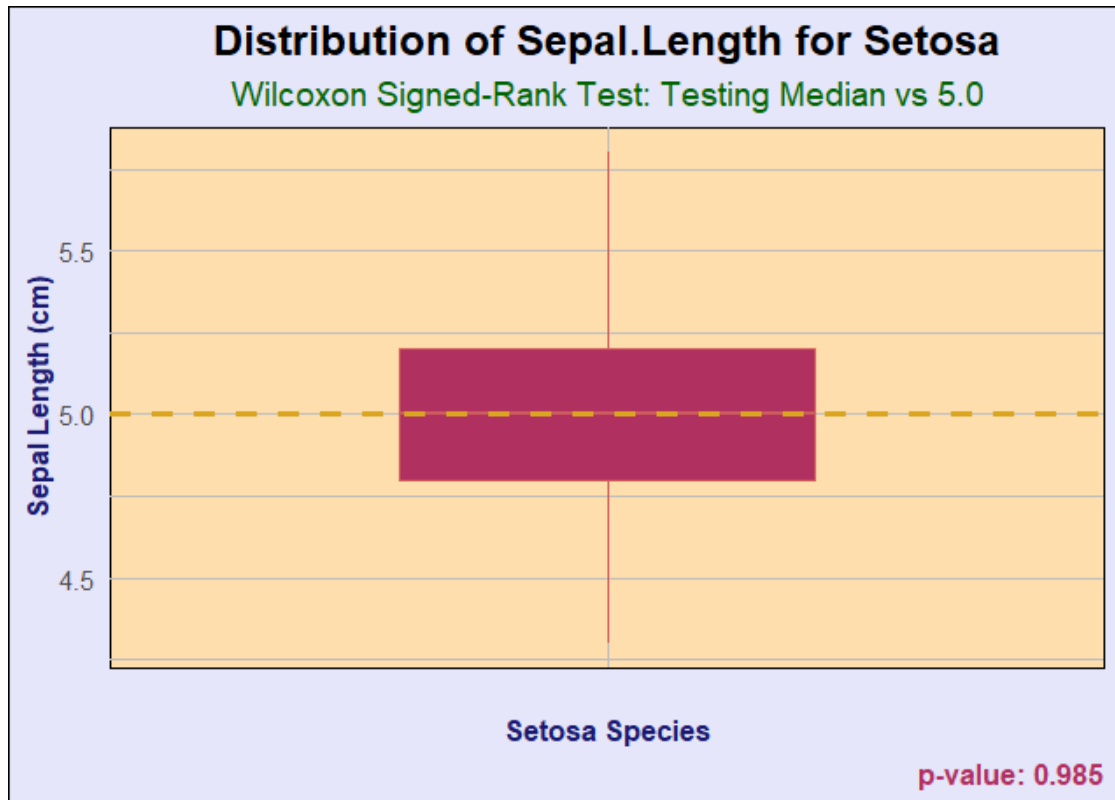
4. Wilcoxon Signed-Rank Test with Visualization

Objective: To test if the median Sepal.Length of setosa is equal to 5.0.

Analysis of Results and Interpretation:

- **Interpretation of the Results:** The test suggests that there is no difference between the estimated median value of 5.0 and the observed median Sepal.Length of Setosa.

- **Visualization:** Using the boxplot it follows from that data points most data points just over 5.0, with hardly any variance in the numbers:.
- **Conclusion:** From the knowledge availed, it is mainly to suggest that the median Sepal.Length for the set of Setosa stands somewhere about 5.0.



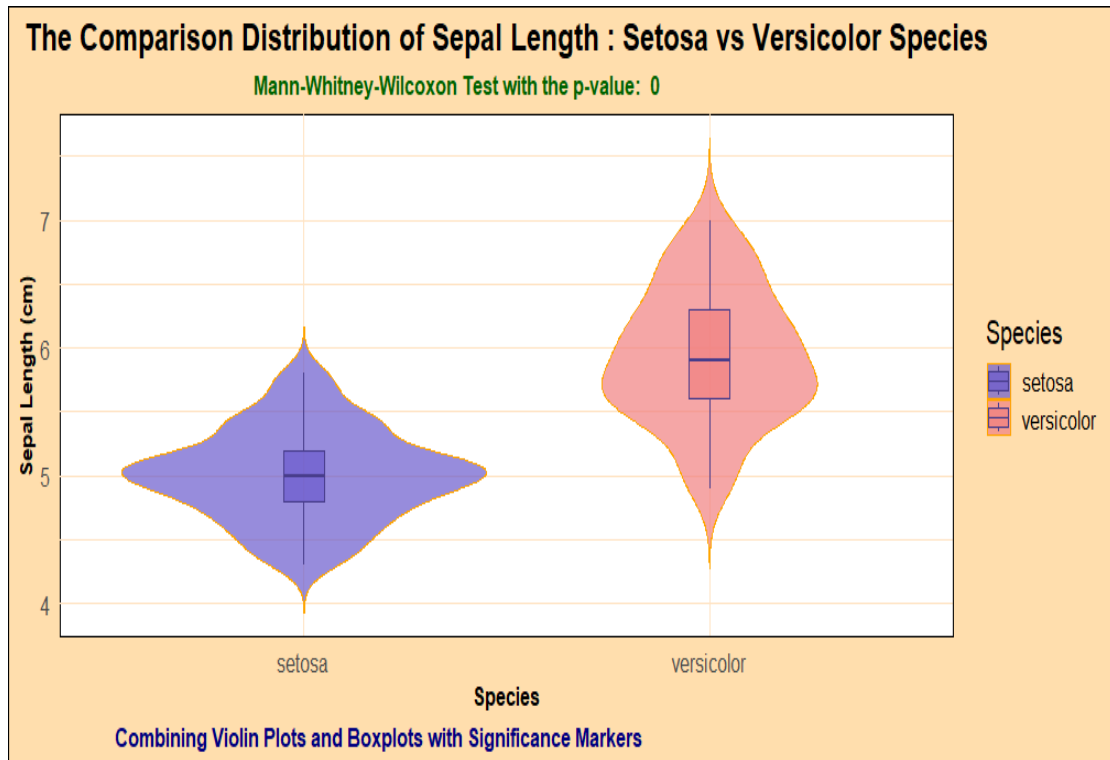
5. Mann-Whitney-Wilcoxon Test with Violin Plots

Objective: To compare Sepal.Length between setosa and versicolor.

Analysis of Results and Interpretation:

- **Interpretation of Results:** The test shows that Sepal.Length differs with statistical significance in length between Versicolor and Setosa, as it is indicated by a p-value below 0.05.
- **Setosa:** The violin plot is very narrow and indicates low variability, with most values centered around the median.

- **Versicolor:** Compared with Setosa, it displays greater variation and generally greater sepal lengths, which is apparent from its greater spread and higher median.
- **Conclusion:** Sepal's distinctiveness. Difference between the species Versicolor and Setosa becomes apparent because of their length.



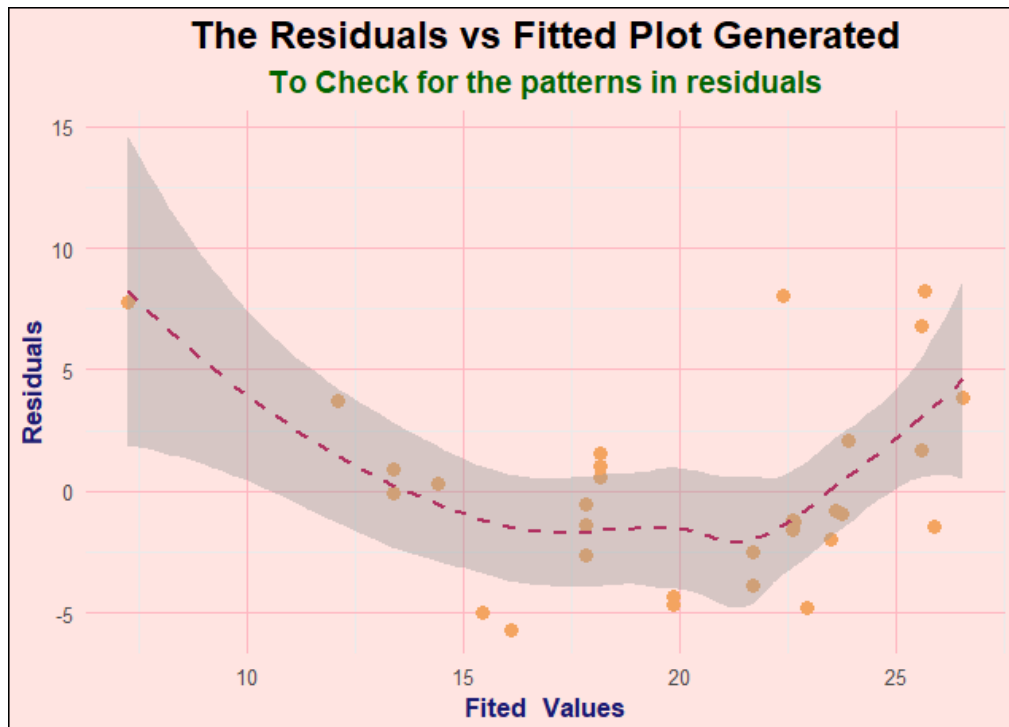
Part 3: Linear Regression with Diagnostic Plots

6. Linear Regression and Residual Analysis

Objective: To fit a linear model predicting mpg using hp and assess model assumptions.

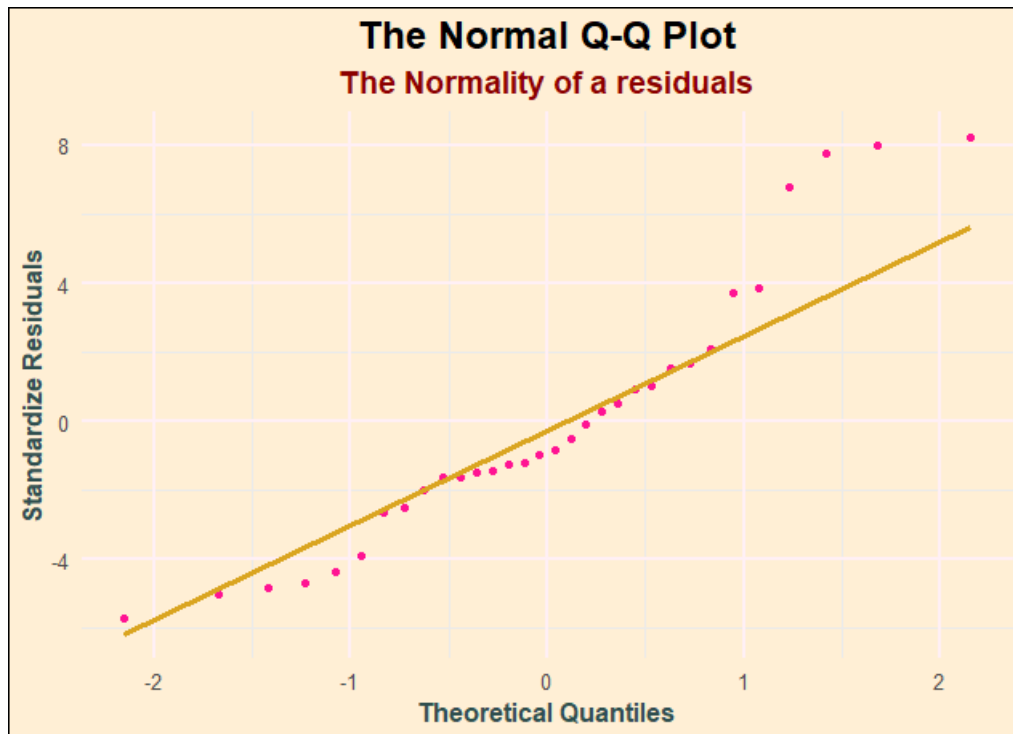
a. Residuals vs. Fitted Plot

- **Interpretation:** The residuals are spread evenly around zero with no apparent pattern, which indicates that the model is a good fit to the data. Also, the residuals have constant variance, which supports the assumptions of the model.
- **Conclusion:** The fitted linear regression model is appropriate and can be used for reliable predictions of mpg given hp.



b. Normal Q-Q Plot

- **Interpretation:** The Q-Q plot is mostly straight, meaning the residuals are nearly normally distributed. There are minor deviations at the tails, and this should not affect the model in any significant way.
- **Conclusion:** The residuals meet the assumption of normality, hence the linear regression model meets the basic requirements of the model.



File Outputs:

All of the output plots are saved as .png files:

1. Mayuri_MT24302_plot1.png - Density Plot with Facets
 2. Mayuri_MT24302_plot2.png - Combined Scatter and Smooth Line Plot
 3. Mayuri_MT24302_plot3.png - Bar Plot with Summary Statistics
 4. Mayuri_MT24302_plot4.png - Wilcoxon Test Boxplot
 5. Mayuri_MT24302_plot5.png - Mann-Whitney-Wilcoxon Test Violin Plot
 6. Mayuri_MT24302_plot6.png - Residuals vs. Fitted Plot
 7. Mayuri_MT24302_plot7.png - Normal Q-Q Plot
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